FREQUENCY ALONE DOES NOT DETERMINE AFFINE RELATION GEOMETRY IN TRANSFORMERS

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ABSTRACT

Prior work finds that relations expressed more often in pretraining data are more likely to admit linear decoders in language-model representations. Does exposure frequency itself determine this geometry? We separate three properties that are often conflated: recalling a relation, representing it with an affine map, and executing it with affine computation. We then construct two controlled synthetic settings. In the first, a one-layer transformer sees each attribute-defined relation fact 16 times. It reaches 0.635 recall on subjects absent from relation training, while an ordinary least-squares (OLS) map fitted on trained subjects selects the correct held-out object with 0.990 accuracy. A frequency-matched random relation remains at chance under the same probe. In the second, an eight-layer GPT-style transformer memorizes a random symmetric relation sampled ten times as often as each of ten attribute-equality controls during joint training. Random-relation accuracy is 0.998, yet the best overdetermined affine fit on all learned entity rows explains only 0.383 of target variance, compared with a mean per-seed minimum of 0.996 for the structured controls. Component interventions show that the forward computation requires content-dependent attention and GELU MLPs. These results give controlled counterexamples to a frequency-only account. Attribute structure, training signals, and the chosen representation determine whether learned relations admit affine descriptions.

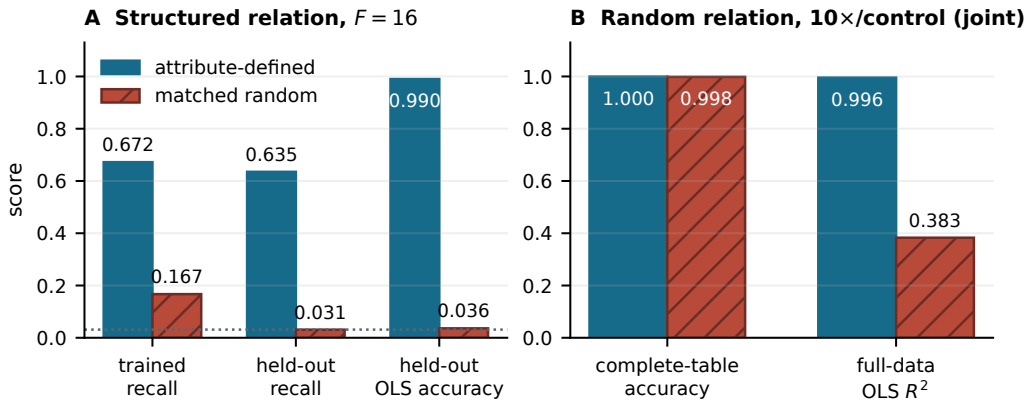


Figure 1: Frequency does not determine affine relation geometry in the two controlled settings. **A:** At 16 occurrences per trained triple, the attribute-defined relation generalizes and an OLS map transfers to subjects excluded from relation training. The frequency-matched random control does not. Chance is $1/32$ for recall and OLS accuracy. **B:** A random relation sampled ten times as often as each structured control during joint training is memorized almost perfectly, but its full-data affine R^2 remains far below every structured relation. Blue solid bars denote structured relations, and red hatched bars denote random controls. Bars report means over three seeds; the structured bars in B report the minimum relation score. The OLS metrics differ by design: held-out nearest-object accuracy in A and overdetermined full-data R^2 in B. Neither metric asserts that the transformer's forward computation is affine.

1 INTRODUCTION

Transformer language models often encode a relation between entities so that one affine transformation predicts the related entity. This observation has supported linear relational decoding and mechanistic accounts of factual recall (Hernandez et al., 2024; Merullo et al., 2025; Valer et al., 2026). One striking empirical result is that affine structure correlates with how often a relation appears in pretraining data (Merullo et al., 2025). Frequency could therefore be a causal driver of relational geometry. It could also be a proxy for structure in the facts, shared attributes, or the supervision induced by the rest of the corpus.

We test the frequency-only explanation in a setting where these factors can be controlled. The key difficulty is definitional. A model can answer a relation correctly without storing it in an affine map. Conversely, an affine map can decode entity embeddings even when the trained forward pass is nonlinear. We therefore measure behavior, representation geometry, and executed computation separately. Figure 1 previews the resulting dissociation.

Our first experiment asks whether a relation with limited direct exposure can acquire shared affine geometry. A one-layer causal transformer learns attributes for 192 subjects and 32 objects. It also sees an attribute-defined relation and a frequency-matched random relation. Subjects excluded from all relation triples provide a genuine transfer test. Attribute training induces an affine-decodable entity geometry before the semantic relation is recalled reliably. Once recall emerges, the same affine map transfers to the excluded subjects.

Our second experiment asks whether high exposure forces a learned relation to be affine. A standard eight-layer decoder learns ten binary attributes, ten equality relations, and one symmetric random lookup table. The random relation receives ten times the examples of any single structured relation during joint training and about five times as many over the full curriculum. The model retains all structured tasks and memorizes the complete random table. An affine map nevertheless leaves most random-relation target variation unexplained on the learned rows themselves. This full-data test avoids an invalid inference from failure to generalize an arbitrary lookup table.

The experiments support three conclusions:

- Low direct exposure is compatible with affine representation geometry when other training facts organize the entities along the relevant attributes.
- High exposure is compatible with a non-affine representation of a relation even in a standard decoder-only transformer with fully trainable embeddings.
- Affine decodability and affine computation are different claims. The first experiment has a strong affine approximation but nonconstant Jacobians, while the second requires nonlinear MLP and attention components for accurate recall.

The result does not dispute a correlation between corpus frequency and linear decodability in natural language models. It shows that frequency cannot, by itself, explain that correlation.

2 THREE LEVELS OF RELATIONAL LINEARITY

Let s denote a subject, r a relation, and o an object. We distinguish three questions.

Behavior. Does the model output the correct object or relation label? We report recall or classification accuracy on the relevant table. Behavior is required for any claim about a learned relation, but it says nothing about how the relation is stored.

Representation geometry. Let $e(s) \in \mathbb{R}^d$ be a specified subject representation and $t_r(s)$ a specified representation of its relational target. We ask whether one affine map fits the entity pairs:

$$t_r(s) \approx M_r e(s) + b_r. \tag{1}$$

The choice of e and t_r is part of the claim. Experiment 1 maps input subject embeddings to output object embeddings and tests the map on subjects excluded from relation training. Experiment 2 maps raw tied subject embeddings to the centroid of positive object embeddings and measures residual error on all learned rows.

Executed computation. A transformer's relation-conditioned state transition is affine only if it has a constant Jacobian and preserves affine combinations on the tested domain. We test midpoint defects and exact-Jacobian dispersion in Experiment 1. In Experiment 2, we intervene on GELU activations and attention routing. A high affine probe score does not imply an affine forward pass, and a nonlinear forward pass does not by itself prove that a particular representation lacks affine geometry.

This separation matters for arbitrary relations. A random lookup has no rule for unseen entries. Poor performance by a probe on held-out random pairs is therefore expected even if the learned entries have a simple representation. Our primary non-affinity test for the random relation instead fits OLS to all 64 learned rows. With 20 input dimensions and an intercept, each target coordinate has 21 coefficients and 64 observations, so the regression is overdetermined.

3 Related work

Linear relational representations. Hernandez et al. (2024) show that affine transformations decode a range of relations in transformer language models. Marconato et al. (2025) study when linear properties of next-token predictors are identifiable, while Valer et al. (2026) evaluate relational linearity empirically under several definitions. These works motivate our use of an explicit map between subject and target representations. Our contribution is to manipulate relation structure and frequency independently, then compare affine geometry with the model's executed computation.

Frequency and relational geometry. Merullo et al. (2025) find a strong association between pretraining-data frequency and linear relational representations, with linear structure appearing consistently but not exclusively at higher frequencies. Their result concerns naturally occurring relations in pretrained models. We test a different question: whether changing exposure is sufficient to determine geometry when data structure and training signals are controlled. Our synthetic counterexamples are compatible with their observed correlation, but exclude a frequency-only causal account.

Factual recall as memory and computation. Feed-forward layers have been described as key-value memories (Geva et al., 2021), and causal interventions localize factual associations to specific transformer components (Meng et al., 2022; Geva et al., 2023). Detailed circuit studies find distributed interaction between entity representations, attention, and MLPs (Nanda et al., 2023). Recent theory and experiments connect associative memory to factual recall (Nichani et al., 2025), analyze geometric mechanisms (Ravfogel et al., 2026), and formalize tradeoffs between relational generalization and memorization (Cheng & Lippl, 2026). We use probes only for representation claims and causal interventions for claims about the forward computation.

Decodability and executed algorithms. A transformer can itself implement regression-like computation: Von Oswald et al. (2023) construct a linear self-attention layer whose data transformation is equivalent to a gradient step on a regression loss. That is a claim about the algorithm executed during a forward pass, whereas our OLS maps are post hoc tests of a named representation. The distinction also appears in controlled formal-language models. Sharma et al. (2026) find that several stack variables are linearly decodable from Dyck-language transformers, but interventions assign them different causal roles. We therefore pair decodability measurements with direct tests or interventions on the trained computation.

4 Controlled experimental design

Table 1 summarizes the two settings. Both use synthetic tokens and ordinary next-token cross-entropy. All parameters, including token embeddings, remain trainable. We report means and sample standard deviations across seeds 19, 31, and 47.

The affine metrics differ deliberately. Experiment 1 asks whether shared geometry transfers to subjects absent from relation training, so held-out nearest-object accuracy is the relevant test. Experiment 2 uses a finite random table with no rule for unseen entries. Its primary test therefore asks whether one overdetermined affine map reconstructs all rows that the model has memorized, which is measured

	Experiment 1	Experiment 2
Question	low-exposure affine geometry	frequent non-affine relation
Entities	192 subjects, 32 objects	64 entities
Structure	2 categorical attributes	10 balanced binary attributes
Relations	semantic + matched random	10 equality + symmetric random
Model	1 block, width 64	8 blocks, width 20
Parameters	84,288	41,760
Embedding setup	untied, trainable	tied, trainable
Primary affine test	held-out OLS accuracy	full-data OLS R^2

Table 1: The two experiments target different directions of the proposed frequency effect. Direct metric values should be compared within each experiment, not across their different target representations.

by full-data R^2. Numerical values should be compared within an experiment, not between these two targets.

4.1 Experiment 1: attribute-defined and random relations

Each subject a_i has attributes $(f_{i \bmod 32}, g_{\lfloor i/32 \rfloor})$, so the 192 subjects cover all 32×6 combinations. Each object b_j has a shifted f attribute. The semantic relation is $R_{\text{sem}}(a_i) = b_{i \bmod 32}$. The control $R_{\text{rnd}}(a_i) = b_{\rho(i)}$ uses a seed-dependent, attribute-independent balanced assignment. For each f value, four subjects enter relation training and two are held out. Each relation therefore has 128 trained and 64 held-out subjects. Every object is the target of four trained and two held-out subjects under either relation.

Attribute sentences describe every subject and object. Each attribute sentence is repeated $A = 128$ times. Each of the 128 triples per relation is repeated F times, where $F \in \{1, 4, 16, 64\}$. At the selected $F = 16$ setting, the corpus has 28,672 attribute sentences and 4,096 relation sentences. One epoch is exactly one shuffled pass through those 32,768 sentences, or 512 batch-64 optimizer updates.

The causal transformer has one block, width 64, four heads, context length 8, and untied input and output embeddings. All 84,288 parameters are initialized from $\mathcal{N}(0, 0.02^2)$. Training uses unweighted language-model cross-entropy and AdamW with learning rate 10^{-3}, five warmup steps, no weight decay, and gradient clipping at 1.

For the representation test, OLS maps each subject's input embedding to the output embedding of its target object. We fit on 128 relation-training pairs. On each of the 64 held-out subjects, we select the nearest of 32 object rows by cosine similarity. The chance accuracy is $1/32$.

4.2 Experiment 2: equality relations and a random table

The second setting contains 64 entities and ten explicit binary attributes. Every attribute column has 32 zeros and 32 ones, all 64 attribute rows are unique, and the centered attribute matrix has rank ten. Relation r_i returns `y` exactly when its two entities agree on attribute i:

$$y_i(s, o) = \mathbb{1}[a_i(s) = a_i(o)]. \tag{2}$$

Each structured table is symmetric, contains the diagonal, and has 32 positive and 32 negative objects per subject.

The eleventh relation, R_{rnd}, is a random symmetric table. We sample a simple 31-regular graph, randomize it with degree-preserving edge swaps, and add self loops. This matches the structured tables' symmetry, diagonal, degree, and class balance. We reject any draw equal to a structured table. Relation prompts take the form `<bos> relation subject object` and predict `y` or `n`. Evaluation enumerates all $11 \times 64^2 = 45{,}056$ ordered prompts. Training uses the same 1:9 positive-to-negative sample ratio for every relation and gives positive labels weight 9.

The model is a decoder-only transformer with eight pre-LayerNorm blocks, residual width 20, four attention heads, GELU MLP width 80, learned token and position embeddings, final LayerNorm, and a tied embedding-output projection. It has 41,760 parameters. There is no bilinear head, relation classifier, direct embedding loss, or frozen parameter.

F	Held-out OLS accuracy R_{sem}	Held-out OLS accuracy R_{rnd}	Trained recall R_{sem}	Trained recall R_{rnd}	Held-out recall R_{sem}	Held-out recall R_{rnd}
1	$1.000 \pm .000$	$.031 \pm .013$	.029	.010	.016	.010
4	$1.000 \pm .000$	$.016 \pm .013$	.055	.034	.057	.031
16	$.990 \pm .007$	$.036 \pm .007$	.672	.167	.635	.031

Table 2: Experiment 1 results, mean $\pm$ standard deviation over three seeds. F is the number of occurrences of each trained relation triple. OLS is fitted on the 128 trained subjects and evaluated on the 64 subjects excluded from all relation triples.

The training corpus includes forward attribute examples and inverse examples of the form `<bos> attribute 0/1` followed by a matching subject. The inverse examples use ordinary tied-vocabulary cross-entropy with weight 4. They provide a task-specific signal that organizes the raw subject rows by attribute; they do not penalize embedding coordinates directly. After 6,000 structured-only steps, joint training samples R_{rnd} for half of each relation batch and splits the other half across the ten equality relations. The random relation is therefore ten times as frequent as any one structured relation during this stage. Including the structured-only stage, it receives about five times as many relation examples over the full curriculum. Joint training runs for 12,000 steps at learning rate 10^{-3}, followed by at most 300 lower-rate continuation steps. Appendix A gives the complete protocol.

For relation r, the affine target is the centroid of positive object embeddings:

$$c_r(s) = \frac{1}{32} \sum_{o: y_r(s,o)=1} e(o). \tag{3}$$

OLS fits $c_r(s) \approx M_r e(s) + b_r$ on all 64 entities. We report multivariate R^2. The same representation, target, intercept, and solver are used for all eleven relations. The centroid encodes the complete positive-object set as one vector per subject and gives the equality relations an analytic affine target. A linear classifier on concatenated subject and object rows would instead test an additive pair score, which cannot express equality without interaction terms and would not provide the intended structured affine control.

5 Low exposure is compatible with affine geometry

At $F = 16$, R_{sem} reaches 0.672 recall on trained subjects and 0.635 on held-out subjects (Table 2). The held-out OLS map selects the correct object with 0.990 ± 0.007 accuracy. The frequency-matched random relation reaches only 0.167 trained recall, chance held-out recall, and chance OLS transfer. The semantic relation has structure that both the model and the affine map can reuse; frequency matching alone does not provide it.

The affine geometry appears before reliable relation behavior. At $F = 1$, OLS transfer is perfect even though semantic recall is near chance. Varying the attribute-sentence repetition A from 2 to 24 produces a transition in OLS accuracy from 0.05 to 1.00 (Figure 2). A control model trained with no relation triples follows the same transition. Attribute facts, rather than the single relation exposure, therefore create the affine-decodable geometry.

Strong affine geometry does not make the executed lookup affine. For a fixed relation token, we fit affine approximations to six learned maps and compute normalized midpoint defects and dispersion of exact Jacobians. The full subject-to-logit map has cross-validated affine $R^2 = .965 \pm .010$, yet its midpoint defect is $.352 \pm .011$ and Jacobian dispersion is $.281 \pm .013$. Dispersion on the empirical subject-difference subspace is $.284 \pm .013$. An exactly affine map has zero defect and constant Jacobian. The trained lookup is thus closely approximated by an affine map on these examples but is not itself affine or Jacobian-equivalent.

Causal analysis identifies how the nonlinear model uses the shared representation. A centroid classifier trained on the 128 relation-training embeddings recovers the f class of all 64 held-out embeddings with perfect accuracy. Swapping the subject-dependent attention value payload between prompts recovers $.987 \pm .008$ of the target-logit change; swapping only attention weights recovers

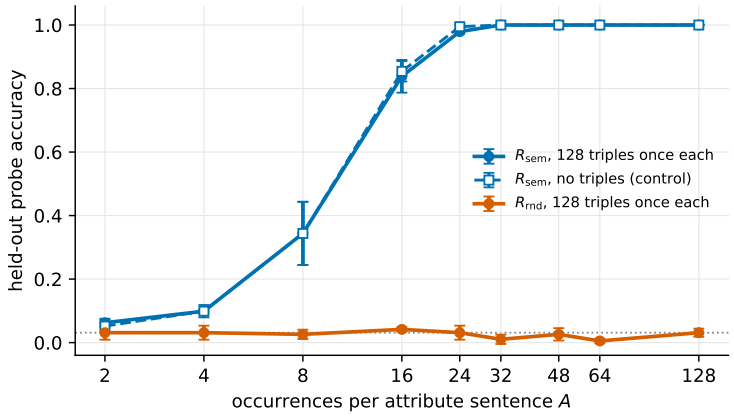


Figure 2: Attribute training creates the affine-decodability transition in Experiment 1. Each semantic triple occurs once in the solid curve; the dashed curve contains no relation triples. Both use a fixed 128-pair OLS fit and 64 held-out subjects. Curves show means over three seeds with one sample-standard-deviation band.

Seed	Min. structured acc.	Random acc.	Min. structured R^2	Random R^2
19	.99951	.99780	.99595	.33058
31	.99927	.99805	.99642	.40353
47	.99976	.99951	.99468	.41460
Mean	.99951	.99845	.99568	.38291

Table 3: Complete-table behavior and full-data raw-embedding OLS in Experiment 2. The structured columns report the minimum over ten equality relations. OLS fits all 64 subject rows, including every row whose relation behavior is evaluated.

$.013 \pm .007$. Replacing all head outputs by their training-subject means reduces object accuracy from $.729 \pm .094$ to chance. A relation-token patch shows that most relation-specific selection occurs in the distributed MLP population. The mechanism transports an attribute-class code through attention and converts it to an object direction through a nonlinear but accurately affine-approximable readout.

6 High exposure is compatible with non-affine geometry

The GPT-style model learns all tasks together. Across the three seeds, attribute accuracy is 1.000, the minimum structured-relation accuracy is 0.99951 ± 0.00024, and random-relation accuracy is 0.99845 ± 0.00092 (Table 3). The minimum prediction symmetry over all eleven relations is 0.99691 ± 0.00185. The comparison therefore does not trade random-table memorization for loss of the structured facts.

The representation geometries differ sharply despite the behavioral match. Across all 30 structured fits, mean full-data R^2 is .99692 and the smallest individual fit is .99468. The three random-relation fits explain .331, .404, and .415 of the target variance. Their relative reconstruction errors are .818, .772, and .765. An affine map therefore fails to approximate most of the learned random centroid variation, even though the transformer predicts the underlying table almost perfectly.

The structured control has an analytic affine target. Writing $b_i(s) = 2a_i(s) - 1$, let μ_i^+ and μ_i^- be the object-embedding centroids for the two values of attribute i. Then

$$c_i(s) = \mu_i^- + \frac{b_i(s) + 1}{2}(\mu_i^+ - \mu_i^-). \tag{4}$$

The inverse attribute examples align b_i with the raw entity rows, allowing OLS to recover this map. The random graph has the same observable table statistics but no corresponding attribute rule.

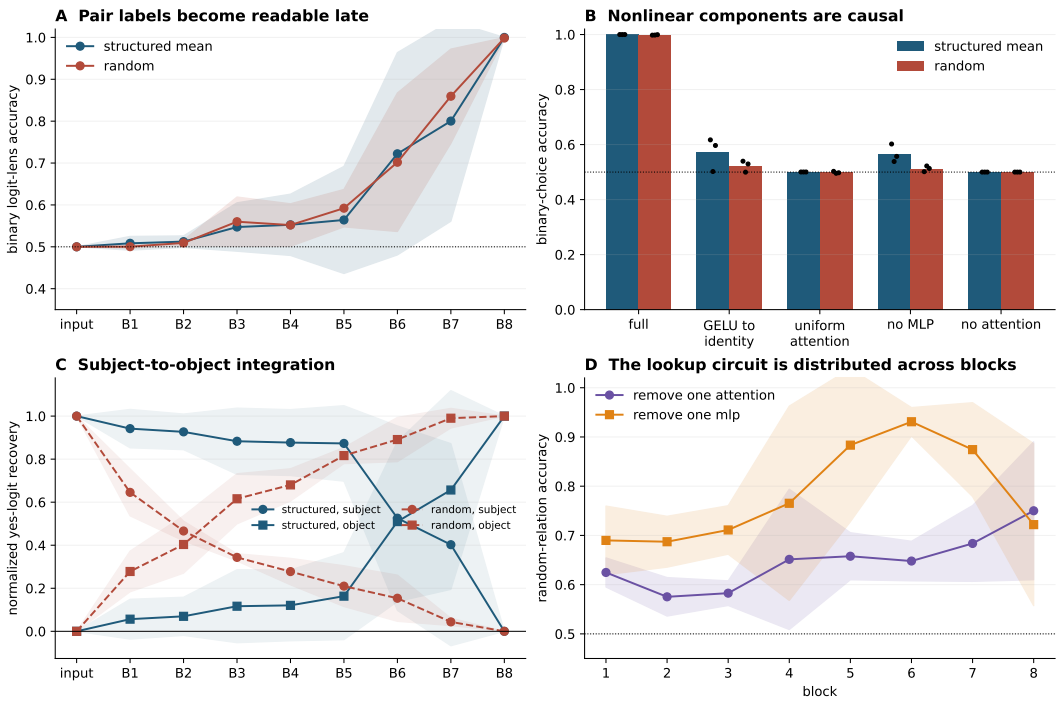


Figure 3: The frequent random relation is implemented by distributed transformer computation. **A:** A logit lens reads the random answer mainly in the last two blocks. **B:** Global interventions on GELU, attention routing, MLP outputs, or attention outputs damage both relation types. **C:** Residual patching shows pair information moving from the subject position to the object position. **D:** Removing any one attention or MLP sublayer impairs random recall, but no single layer contains the entire lookup. Lines and bars show means over three seeds; bands and error bars show one sample standard deviation.

The random table retains a partial affine trace. A simultaneous row-column permutation null has mean R^2 near .338. The random score lies at the 38th null percentile for seed 19 and above the 98th percentile for seeds 31 and 47. We do not claim that its affine component is zero. The relevant separation is between a nearly exact behavioral lookup, $R^2 = .383$ for its best raw-embedding affine map, and $R^2 \geq .995$ for every structured control.

Four-fold cross-validated random R^2 is negative, but this is not evidence for our non-affinity claim. A held-out portion of a random table cannot be inferred by either a transformer or an affine probe. The primary full-data residual instead asks whether one affine map can reconstruct the rows that the trained model has already learned. Appendix C reports cross-validation only as a fit-stability diagnostic.

7 The frequent random lookup uses distributed nonlinear computation

We modify the forward pass of each trained model while keeping its weights fixed. Replacing every GELU by the identity reduces random-relation accuracy from .998 to $.523 \pm .021$. Uniform causal attention gives $.499 \pm .004$, removing all MLP outputs gives $.512 \pm .010$, and removing all attention outputs gives .500. The learned affine weights and residual additions remain unchanged in the GELU intervention, so nonlinear activation is causally necessary for the trained computation.

Layerwise diagnostics locate the computation (Figure 3). A logit lens remains near chance through block 2, reaches .560 after block 3, .702 after block 6, .860 after block 7, and .998 after block 8. Removing one attention sublayer at a time lowers random accuracy to .575–.750; removing one MLP sublayer gives .687–.931. The table is not localized to one block.

Residual-stream patching tracks where pair information travels. At successive block boundaries, patching the clean subject state into a corrupted prompt recovers 1.000, .646, .466, .344, .278, .210, .154, .044, and .000 of the clean answer margin. Patching the clean object-position state recovers .000, .278, .403, .616, .680, .817, .891, .990, and 1.000. Attention moves subject and relation information to the object position, while the later GELU MLPs transform the combined state into the final `y`/`n` margin.

The interventions also damage the structured relations. They establish that the trained forward computation uses nonlinear components, but they do not by themselves distinguish how the two relation types are represented. The full-data raw-embedding OLS comparison supplies that relation-specific evidence.

8 Discussion and limitations

Frequency is a correlate, not a complete mechanism. The two experiments reverse the expected association in controlled settings. In Experiment 1, attribute supervision creates an affine entity geometry with little direct relation exposure. In Experiment 2, repeated sampling lets a transformer memorize an arbitrary relation without bringing its raw-embedding geometry close to the structured affine controls. Natural corpora can couple frequency to many other properties, including shared attributes, paraphrase diversity, and entity coverage. Any causal account of observed relational linearity must model those signals as well.

The representation must be named. Our conclusions concern explicit maps. Experiment 1 uses input subject embeddings and output object rows. Experiment 2 uses raw tied entity embeddings and positive object centroids. The results do not rule out an affine random-relation feature at some hidden layer or under a different target construction. They do show that the proposed raw subject-to-object map cannot approximate the learned random table at the level achieved by the matched structured controls. We did not conduct a systematic affine-probe search over intermediate layers, so the claim should not be read as layer-independent.

Representation and computation are distinct. Experiment 1 has excellent affine transfer while direct midpoint and Jacobian tests reject exact affinity of the lookup. Experiment 2 combines a poor raw-embedding affine fit with causal dependence on GELU and content-dependent attention. These measurements answer different questions. Probe performance alone should not be used to label an entire circuit linear or nonlinear.

The random relation tests memory, not generalization. The model repeatedly samples a finite random graph and evaluation enumerates that graph. There is no latent rule for unseen random edges. This setting isolates how a known table can be stored, but it does not model systematic generalization to new entities or facts.

The curriculum is specialized. The inverse attribute examples in Experiment 2 deliberately organize the tied raw embeddings. They are standard next-token examples rather than a direct embedding penalty, but generic language-model pretraining need not supply an equivalent signal. We did not ablate these examples after selecting the final configuration; the near-perfect structured raw-embedding fits are therefore conditional on this curriculum. Whether the random relation's raw-embedding R^2 changes without that signal remains untested. The architecture is a compact successful configuration, not a proof of global minimality. Larger models, natural text, other relation arities, and hidden representations remain open settings.

The experiments establish existence, not a frequency curve. Occurrence counts are meaningful within each controlled corpus but are not directly comparable across the two tokenizations and optimization schedules. The study shows that low exposure can coexist with affine geometry and that high relative exposure can coexist with non-affine geometry. It does not estimate a monotonic response to frequency. Each configuration was selected using behavioral and geometry gates and evaluated over three seeds. The large within-setting separations are stable across those seeds, but broader architecture and seed sweeps would be needed to estimate how often each outcome occurs.

9 Conclusion

Exposure frequency alone does not determine whether a transformer relation admits an affine representation. Shared attribute training can create affine geometry before a relation is recalled, while a highly sampled arbitrary relation can be memorized without a comparably accurate affine map in raw entity space. Careful claims about relational linearity should state the behavior being learned, the representations connected by the proposed map, and whether the evidence concerns a decoder or the executed computation.

AI use statement

Generative AI tools were used to help design and critique the synthetic experiments, implement and edit software, generate synthetic data and plots, analyze and interpret experimental outputs, search and summarize literature, and draft and edit this manuscript. The authors inspected the generated code, ran the reported experiments and automated tests, checked numerical claims against archived result files, reviewed the cited primary sources, and revised the resulting text. The authors take responsibility for the paper's claims and contents.

Reproducibility statement

Sections 4–7 define the datasets, architectures, training curricula, metrics, and interventions. The appendix gives optimization details, per-seed results, and the complete affine tests. The supplementary code contains deterministic data generators, saved configurations, result manifests, and figure scripts for seeds 19, 31, and 47.

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A Complete training protocols

A.1 Experiment 1

Every training sentence contains five tokens. Attribute sentences have the form `x subject f-attribute g-attribute x`; relation sentences have the form `x subject relation object x`. The vocabulary contains 265 tokens: one separator, two relations, 192 subjects, 32 objects, 32 f values, and six g values. Object b_j has attributes $(f_{(j+3) \bmod 32}, g_0)$, while the semantic target of a_i is $b_{i \bmod 32}$. Thus semantic targets are linked by a consistent attribute shift rather than string identity.

For each f value, a seed-dependent split assigns four subjects to relation training and two to evaluation. The random target function is balanced within both groups. Under either relation, every object is the target of four training subjects and two held-out subjects. The relation frequencies, subject frequencies, and object-target frequencies therefore match exactly.

Training runs for one complete shuffled pass through the corpus. At batch size 64, the number of optimizer updates is 452 for $F = 1$, 464 for $F = 4$, 512 for $F = 16$, and 704 for $F = 64$. The objective is unweighted next-token cross-entropy at every position. Data assignment, model initialization, and shuffle order vary with the seed.

A.2 Experiment 2

The vocabulary has 90 tokens: one beginning token, 64 subjects, ten attributes, eleven relations, and output symbols `0`, `1`, `n`, and `y`. Context length is four. Each pre-LayerNorm block contains causal multi-head self-attention and a four-times-wider GELU MLP with residual connections. Head width is five. Token embeddings, position embeddings, eight blocks, final LayerNorm, and the tied output projection are trainable throughout.

Training uses AdamW with no weight decay, gradient clipping at 1, and a 100-step linear warmup. Relation batches contain 1,024 examples; attribute batches contain all 640 facts. Forward attribute loss has weight 1 and inverse attribute loss has weight 4. The ten equality relations train with both attribute directions for 6,000 steps at learning rate 10^{-3}. Joint training then samples the random

relation for half of each relation batch and splits the remainder uniformly among equality relations for 12,000 steps at 10^{-3}. Seed 47 uses 200 continuation steps at 3×10^{-4}; seeds 19 and 31 use those steps plus 100 steps at 10^{-4}. The selected checkpoints are the first evaluated checkpoints satisfying the behavioral, symmetry, and structured-geometry gates.

B Additional Experiment 1 results

Map	Affine R^2	Relative error	Affine decision	Midpoint defect	Jacobian dispersion
Attention core	$.993 \pm .002$	$.080 \pm .012$	$.742 \pm .072$	$.035 \pm .007$	$.085 \pm .013$
Attention block	$.983 \pm .004$	$.129 \pm .015$	$.750 \pm .066$	$.273 \pm .010$	$.179 \pm .005$
MLP core	$.983 \pm .003$	$.131 \pm .011$	$.742 \pm .061$	$.095 \pm .005$	$.137 \pm .007$
MLP block	$.974 \pm .006$	$.160 \pm .017$	$.758 \pm .057$	$.247 \pm .027$	$.172 \pm .009$
Readout	$.993 \pm .003$	$.081 \pm .017$	$.760 \pm .060$	$.199 \pm .004$	$.228 \pm .007$
Full lookup	$.965 \pm .010$	$.185 \pm .027$	$.750 \pm .056$	$.352 \pm .011$	$.281 \pm .013$

Table 4: Transformation-level tests for R_{sem} at $F = 16$. Affine fits use four target-stratified folds. An exactly affine map has zero midpoint defect and zero Jacobian dispersion. The affine decision column evaluates the classifier after replacing a map output by its affine prediction; decision preservation does not establish exact affinity.

At $F = 64$, the random relation reaches perfect trained-subject recall but remains at chance on held-out random targets and under the held-out affine probe. This confirms that the architecture can store the random training pairs when exposed often enough. It does not generalize a rule that the data do not contain.

The mechanism interventions use target-logit margin recovery. A patch's recovery is the change produced by the patch divided by the clean-minus-corrupted change, aggregated before division. For the semantic relation, patching all subject-derived head value payloads recovers $.987 \pm .008$ of the effect, while patching only attention weights recovers $.013 \pm .007$. Patching the MLP population recovers $.850 \pm .017$ of the relation-token effect. Keeping only the 32 MLP units with the largest causal effects recovers $.498 \pm .029$ of the subject-specific margin, while ablating them still leaves $.790 \pm .032$ of the effect. The readout is distributed.

C Additional Experiment 2 results

Seed	Attribute	Min. struct.	Random	Min. symmetry	Min. struct. R^2	Random R^2	Random CV R^2
19	1.00000	.99951	.99780	.99561	.99595	.33058	$-.78806$
31	1.00000	.99927	.99805	.99609	.99642	.40353	$-.59251$
47	1.00000	.99976	.99951	.99902	.99468	.41460	$-.66663$
Mean	1.00000	.99951	.99845	.99691	.99568	.38291	$-.68240$

Table 5: Complete-table behavior and raw-embedding affine fits. The cross-validated column is a secondary fit-stability diagnostic, not evidence that a random lookup is non-affine.

For the component interventions, structured-relation mean accuracy is 0.99993 ± 0.00005 in the trained model, 0.57194 ± 0.06142 after replacing GELU with the identity, 0.50035 ± 0.00001 with uniform attention, 0.56585 ± 0.03274 with all MLP outputs removed, and exactly .500 with all attention outputs removed. The random-relation values are 0.99845 ± 0.00092, 0.52295 ± 0.02070, 0.49862 ± 0.00366, 0.51229 ± 0.01040, and .500, respectively. The common degradation shows that both relation types use nonlinear transformer components. It does not erase the representation-level OLS separation.

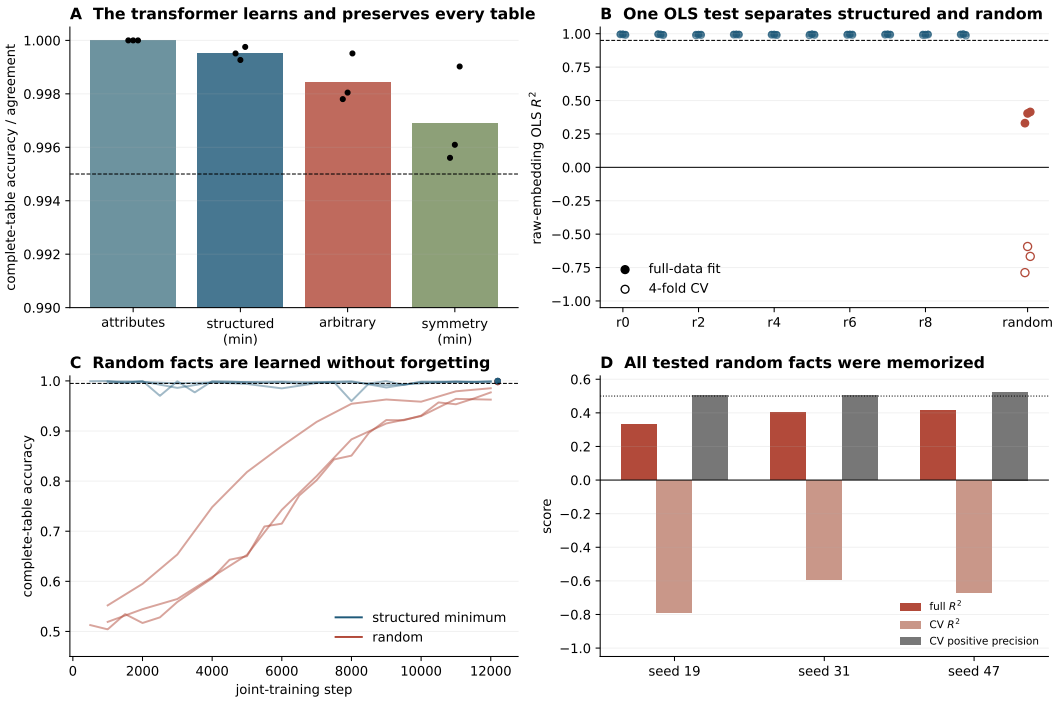


Figure 4: Training and affine-fit diagnostics for Experiment 2. Filled OLS points use all 64 learned rows. Open points show four-fold cross-validation and are included only to diagnose fit stability. Dashed lines mark the prespecified behavioral and structured-geometry gates.

D METRIC DETAILS

For an Experiment 1 stage map h, write $z_i = h(x_i)$. Midpoint defect is

$$D_{\mathrm{mid}} = \frac{\left(\sum_{(i,j)\in\mathcal{P}} \left\| h\left(\frac{x_i+x_j}{2}\right) - \frac{z_i+z_j}{2} \right\|_2^2\right)^{1/2}}{\left(\sum_{(i,j)\in\mathcal{P}} \left\| \frac{z_i-z_j}{2} \right\|_2^2\right)^{1/2}}. \tag{5}$$

The diagnostic fold contains one trained subject for each of the 32 targets, and $\mathcal{P}$ contains all 496 unordered pairs with different targets. We compute each midpoint through the continuous residual-stream extension of the trained network while holding the separator and relation token fixed.

Let $J_i = \partial h/\partial x|_{x_i}$ and $\bar{J} = 32^{-1}\sum_i J_i$. The reported Jacobian dispersion is

$$D_J = \frac{\left(\sum_i \|J_i - \bar{J}\|_F^2\right)^{1/2}}{\left(\sum_i \|J_i\|_F^2\right)^{1/2}}. \tag{6}$$

Jacobians are exact automatic-differentiation Jacobians, not finite-difference estimates, evaluated at all 32 subjects in that fold. Tangent dispersion applies the same formula to J_iQ, where the columns of Q span the empirical subject difference subspace retained at singular values above 10^{-6} of the largest. We use the first target-stratified fold for these diagnostics and replicate the calculation across the three training seeds; we did not resample diagnostic folds.

For Experiment 2, OLS minimizes

$$(\widehat{M}_r, \widehat{b}_r) = \arg\min_{M,b} \sum_{s=0}^{63} \|c_r(s) - Me(s) - b\|_2^2 . \tag{7}$$

The reported coefficient of determination is

$$R_r^2 = 1 - \frac{\sum_s \|c_r(s) - \widehat{M}_r e(s) - \widehat{b}_r\|_2^2}{\sum_s \|c_r(s) - \bar{c}_r\|_2^2}. \tag{8}$$

All output coordinates are pooled in the numerator and denominator. Relative reconstruction error is $\sqrt{1 - R^2}$. Since OLS is the unconstrained minimizer of this full-data residual, ridge regularization or a low-rank constraint cannot increase the primary training-set R^2. They can change cross-validated behavior, which is not the non-affinity test used here. A bilinear head or a linear classifier over explicit subject-object interaction features tests a different representation class rather than another affine map from $e(s)$ to $c_r(s)$.

For residual patching in Experiment 2, a clean prompt uses a correctly classified positive subject and a corrupted prompt uses a correctly classified negative subject while holding relation and object fixed. For answer margin $m = \ell_{\mathrm{y}} - \ell_{\mathrm{n}}$, recovery at residual boundary k is

$$\mathrm{recovery}_k = \frac{\sum_j (m^{(j)}_{\mathrm{patched},k} - m^{(j)}_{\mathrm{corrupt}})}{\sum_j (m^{(j)}_{\mathrm{clean}} - m^{(j)}_{\mathrm{corrupt}})}. \tag{9}$$

Patching copies either the clean subject-position state or the clean object-position state into the corrupted run, then executes all remaining blocks.